

CHENRUI GAO

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EDUCATION

University of Texas at Austin

M.S. in Computer Science

Austin, TX

Aug 2026 – May 2028 (expected)

University of Michigan, Ann Arbor

B.S.E. in Computer Engineering (GPA: 3.93/4.00)

Ann Arbor, MI

Aug 2024 – May 2026

Shanghai Jiao Tong University

B.E. in Mechanical Engineering (GPA: 3.70/4.00)

Shanghai, China

Aug 2022 – Aug 2026 (expected)

PUBLICATIONS

VAMP-MR: Vector-Accelerated Motion Planning and Execution for Multi-Robot-Arms

Philip Huang, **Chenrui Gao**, Jiaoyang Li

IROS 2026

RESEARCH EXPERIENCE

Carnegie Mellon University, ARCS Lab

Research Intern; advised by Prof. Jiaoyang Li — basis of the VAMP-MR publication

Pittsburgh, PA (Remote)

Nov 2024 – Oct 2025

- Built the planner's throughput-critical core in C++: batched forward-kinematics and collision checks across robot configurations into CPU SIMD lanes with a struct-of-arrays layout, removing motion validation as the bottleneck; **10–100×** faster, with millisecond-scale multi-arm motion generation
- Implemented the CBS-MP multi-robot motion planning framework on top of the vectorized primitives

University of California, Irvine

Research Intern; advised by Prof. Sven Koenig — Topological Pathfinding with Ray-Cut CBS

Irvine, CA

Jun 2025 – Aug 2025

- Proposed a hierarchical approach for constrained multi-agent routing: computing homotopy classes over 2D space to guide Conflict-Based Search (CBS)

OPEN-SOURCE CONTRIBUTIONS

verl-omni — diffusion/omni RL post-training | 10 PRs merged

Jul 2026 – Present

- Implemented **On-Policy Distillation** for diffusion RL, the core feature of official roadmap **RFC #293** (PRs #300 and #325, both **merged**): teacher-anchored distillation losses (KL and flow-matching MSE) plus a frozen-teacher runtime that runs as a third forward-only worker alongside actor/ref, with fit-loop hooks and fail-closed config checks, following upstream verl conventions
- Verified it end to end on SD3.5-Medium with an OCR reward: distillation alone lifted student reward from **0.23 to 0.62–0.71**, reaching the FlowGRPO RL teacher's level about **5×** faster than GRPO training, with distillation KL falling monotonically from 4.2e-4 to 1.8e-5; backed by 150+ unit tests (mutation-tested), GPU smoke runs, and a 40-step training-curve reproduction
- Wrote a maintainer-endorsed **RFC** measuring rollout-training numerical consistency across 3 diffusion pipelines (SD3.5-Medium, Qwen-Image-20B, BAGEL-7B) and 8 experiment arms; its proposal shipped as default per-timestep monitoring (#291)
- Traced a replay crash and one-directional log-prob drift to a stale BAGEL step-count mapping (#279); the fix cut mean log-prob drift **157×**, down to bf16 noise, matching an analytical prediction
- Wired the torch profiler into both trainer fit paths and the reward-model rollout server (#252, #256): profiled-step time **616s to 70s**, trace size 163MB to 32MB

sglang-omni — Qwen3-Omni inference performance (official roadmap subtask)

Aug 2026

- Rewrote the thinker-prefill multimodal embedding merge from a per-request Python loop into one batched GPU scatter (PR #1161, **merged** with 2 maintainer approvals): cut device-to-host syncs from **96 to 0** per batch for a **2.1–4.1×** speedup, clearing the path to prefill CUDA graphs
- Verified the rewrite three ways: 6000-seed differential fuzzing (element-wise identical), byte-identical end-to-end serving logits A/B, and an H800 performance matrix

vllm-omni — SANA-Video parallelism & acceleration stack (maintainer-partitioned, issue #5432)

Aug 2026

- Derived a **sequence-parallel scheme for ReLU linear attention**, where existing Ulysses/Ring paths don't apply: since the KV state is a sum over the sequence dimension, SP needs only one all-reduce of a single $d \times d$ state, i.e. $O(d^2)$ communication regardless of sequence length; measured **1.80×** at SP2 and **3.57×** with SP2+CFG2 on 4xA800 (PR #5940, *in review*)
- Added tensor-parallel and CFG-parallel execution: TP2 **1.52×**, CFG2 **2.04×**, TP2×CFG2 **3.02×**, with a 32-case correctness matrix and TP1 bit-exact against baseline (PR #5861, *in review*)

- Added Cache-DiT caching with CPU offload: **1.56×** faster, peak GPU memory down **7.3 GiB** (PR #5882, *in review*)
- Fixed a FLASH_ATTN cross-attention key-padding bug that silently mixed samples within a batch—affecting every padded cross-attention diffusion model—raising real-weight output cosine from **0.9157 to 0.99996** (PR #5866, *in review*); omni stage-runner bugfix **merged** (PR #5269)
- Built a shadow-wrapper merge-on-load fast path for diffusion LoRA (fp32 delta fold-in, exact unload): LoRA generation overhead **+122% to +4.7%**, PPO ratio held at $1\pm 1e-6$ (PR #5059, *in review*)

PROJECTS

LLM Training Systems from Scratch (Stanford CS336 Assignments) | *PyTorch, Triton, NCCL* 2026

- Wrote **FSDP from scratch** (backward-hook reduce-scatter, on-demand all-gather, fp32 master shards, round-robin sharded optimizer), a **FlashAttention Triton kernel**, and a **KV cache**, all inside a Transformer LM built from the ground up (RoPE, BPE tokenizer)

TEACHING

Teaching Assistant, Intro to Computers & Programming (Prof. Jigang Wu) *SJTU, Fall 2023*

HONORS

Dean's Honor List, University of Michigan *Dec 2024, May 2025*
 John Wu and Jane Sun Sunshine Scholarship *Nov 2023*

SKILLS

Languages: C++, Python, Triton, CUDA (basic), MATLAB

ML Systems: PyTorch (torch.distributed, torch.compile), vLLM, verl, FSDP/DDP + LoRA, FlashAttention, Ray, Hydra, torch profiler

Systems & Tools: Linux, Git/GitHub OSS workflow, CUDA 13/cu130, ROS, OpenCV, NumPy, Pandas

Coursework: Stanford CS336 (LLMs from Scratch), MIT 6.S081 (Operating Systems)